

SF-7016 POULTRY STUNNING SYSTEM



INSTALLATION, OPERATION & MAINTENANCE MANUAL

SIMMONS ENGINEERING COMPANY

91 SIMMONS INDUSTRIAL PLACE

P.O. BOX 546

DALLAS, GA 30132

770-445-6085 FAX: 770-443-9058

INFO@SIMENGCO.COM



March 11, 2020



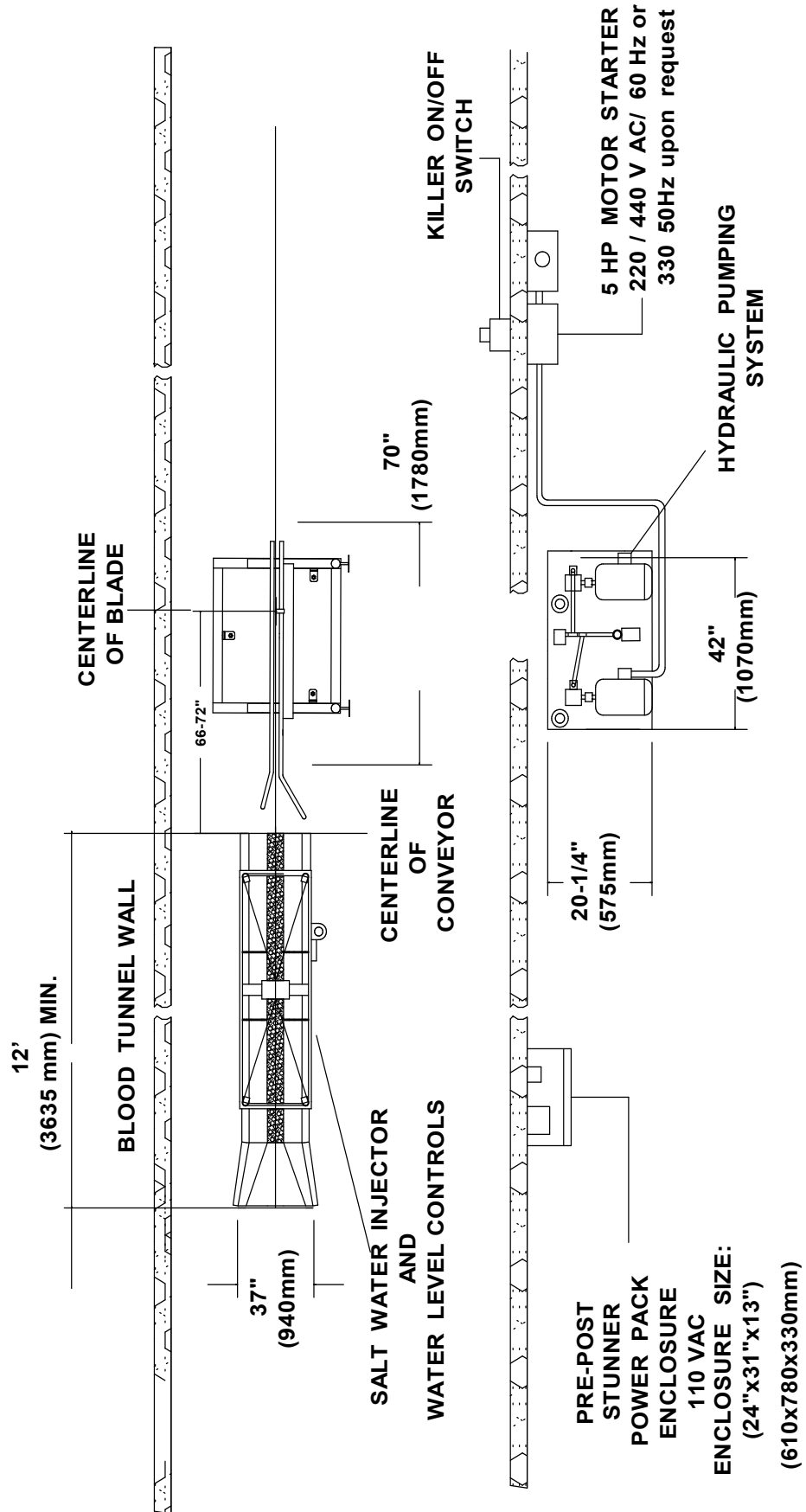
*****FOR THE MOST UP TO DATE MANUALS PLEASE EMAIL
INFO@SIMENGCO.COM*****



Table of Contents

General Information	4 - 7
Introduction - Description of System	5 - 5
System Overview	5 - 6
Safety Warnings	6 - 7
Installation and Operation	8 - 17
Pre-Installation Instructions	8 - 11
Installation and Setup	11 - 14
How to Operate	15 - 17
Maintenance and Cleaning Instructions	18 - 20
Troubleshooting the System	21 - 21
Diagrams	22 - 27

OVERALL DIMENSION
DEPENDS ON BIRD PER HOUR
RATE IN EACH PLANT



General Information

1.1 Introduction

This manual will help you to understand, set up and safely use the Simmons SF-7016 Stunner. It is important to adhere to all safety warnings and cautions that appear throughout this publication for the welfare of the birds as well as the stunner control operator.

1.2 System Overview

The Simmons SF-7016 Stunner has been engineered based upon the practice of using our dual patented combination of frequency and current to accomplish unconsciousness with not only a reversible, simple stun but an irreversible stun when required while sparing any pain. The SF-7016 is used in combination with neck cutting to kill with a sharp knife whether mechanically or manually. When combined with the Simmons SK-5 Killing Machine, the SF-7016 will ensure the most efficient bleed out.

The SF-7016 Stunner is designed to accommodate a range of poultry sizes and species from broilers to turkeys by following the recommended parameters.

The overall system includes a stunner cabinet, controls and overhead support frame, which is attached to a pre-existing overhead track with kill-line shackles. The SF-7016 cabinet is open at each end to allow poultry to easily enter and exit the cabinet. The stunner cabinet has a wide “U”-shaped funnel design to accommodate larger birds as they enter. The SF-7016’s elevated infeed ramp ensures that birds do not experience any pre-stun shocks until contact with the insulated rump, breast bars and saltwater saline solution. The newly designed cabinet includes a saltwater injection system to fill and maintain the appropriate level of saltwater saline solution in the cabinet. A float switch assembly controls the depth and flow of the solution electronically, preventing under or overflow. The use of the salt water saline solution not only increases conductivity but assists in the overall optimization of current flow of the System.

The cabinet includes a DC section, which is located at the entrance end and an AC section located at the exit end of the cabinet. The DC section renders the bird unconscious without damaging effects of severe muscular contractions; while the AC section gives a reversible or irreversible stun based upon system settings. Both sections include Simmons patented contact grate. This patented combination technique prevents bird and meat quality defects that can frequently be caused by stunners which use only high voltage AC.

General Information

Dependent upon line speed, condition of birds and maintenance of the Stunner, birds can be stunned within range of 14-50 volts DC, directly followed by 22-32 volts AC. The resulting current renders the bird unconscious and relaxes each bird to allow its head to be directed into the Simmons Killing machine. When the bird exits the Stunner, the head should be relaxed, and the wings should be “tucked-in” or contracted. As a result, the stunning process is crucial in assuring that the birds will experience no pain or anxiety during the killing process.

The stunner control panel consist of one NEMA 4X stainless steel enclosures containing Simmons power supplies for the DC and AC controls with digital displays. The power supplies operate by supplying a constant current to the stunner. Power supplied to the system is controlled by settings established by a trained operator.

Simmons SF-7016 Stunner is designed to be modular, offering the unique capability of increasing or decreasing sections to the demand of line speeds. The System also offers manual, electric, hydraulic and pneumatic overhead adjustment systems, which allows for adjustment to the height of the cabinet in order to accommodate a variety of species and sizes of poultry.

1.3 Safety Warnings



WARNING! DO NOT ATTEMPT TO USE THIS PRODUCT BEFORE READING AND UNDERSTANDING THESE INSTRUCTIONS! FOR ASSISTANCE CONTACT SIMMONS.



WARNING: Before operating, maintaining or servicing this System, read and understand all the technical data included for your Simmons Systems. This publication contains information that is vital for you to know and understand. Failure to read and understand this publication in its entirety before operation increases the likelihood of inadequate stunning performance and is extremely dangerous for the bird, the System and the operating personnel.



WARNING: This manual **MUST** be read thoroughly by ALL personnel who operate, clean, or maintain this equipment! Special care should be taken to ensure that the **WARNINGS** and safety requirements for operating and servicing the equipment are followed. The user should be aware of and adhere to ALL local building and fire codes, ordinances, and **SAFETY STANDARDS** prior to installing, operating, and/or servicing this system.

General Information

-  **WARNING:** Individuals operating and maintaining the SF-7016 must be properly trained in all aspects of animal welfare, including both the manufacturer's instructions for the processor, and the regulations ordained by all governing bodies concerned. Those individuals should regularly ensure that the SF-7016 is operating properly and maintained in accordance with the manufacturer's recommendations; ensuring that they are in compliance with local standards. These individuals must take immediate action when the stunning system is not operating properly. Without diligently overseeing the equipment, its settings and its maintenance, undo physical harm, discomfort, or anxiety could be caused to the birds.
-  **WARNING:** Before operating, always inspect all settings and cleanliness of the full system. Properly inspecting and maintaining the system can reduce the risk of inadequate stunning performance. Before use, confirm that all parts are functioning properly. Simmons Systems should be maintained and repaired in accordance with the manufacturer's specifications. Only Simmons authorized replacement parts are to be used with this system and should not be modified from the manufacturer's original design. Replace worn or broken parts immediately. Maintenance and recommended standard operating procedures are enclosed.
-  **WARNING:**  Indicates hazardous voltages may be present. Use standard safety precautions to avoid personal contact with these voltages. For maximum safety, DO NOT TOUCH this system while in operation. Do not touch any object that could provide a current path to / from the system. Before performing any maintenance, lock out the System by use of the facility's standard operating procedures.
-  **WARNING:** Before every use, verify that the power supply controllers are set to the manufacturer's recommended settings for the specified stun. This is crucial in assuring that the birds will experience as little pain and anxiety as possible throughout the process. Do not test on a live bird. Verify outputs of electrical loads and stimulate the resistance of the birds with the use of meter.
-  **WARNING:** To reduce the risk of possible electric shock, a qualified electrician should perform all necessary wiring.

Installation and Operation

2.1 Pre-Installation Checklist:

As each installation is unique, this information is intended to provide a general guide for your Simmons System. Consult your Simmons representative for specific directions pertaining to the installation of your system.

SF-7016 Simmons Stunner System includes:

This is a general System overview of items included in the SF-7016 to be used in reference throughout this manual.

- A. Modular fiberglass cabinet assembly with DC and AC Stunning sections
 - a. Simmons patented grate and extension for Alternating Current
 - b. Infeed Section with ramp / Turkey infeed
 - c. Upper cabinet
- B. Stainless steel overhead support assembly with adjustable frame for height control.
- C. Stainless steel or plastic guide bar kit for shackles.
- D. Water level control assembly with saltwater injector system.
- E. NEMA 4x Stainless steel enclosure for controllers with digital display assembly and Primary and standby AC and DC power supplies

2.2. Preparing the Site:

The following instructions are provided to prepare for the installation of the SF-7016 Simmons Stunner ahead of time, thereby minimizing the work to be done during the time of installation.

2.2.1 Layout:

The condition of the live bird prior to stunning and killing is of the utmost importance. The best possible layout of the SF-7016 Stunner System and killer should be set up in a straight, unobstructed line directly from the live hang area to the killing room. All efforts should be made to limit the number of corners and track elevations as well as the entry ramp and approach of the shackle line for the bird. With consideration to the overall line speed, the shackle line should be kept to a minimum; while allowing adequate time for the birds to calm by the time they reach the SF-7016 System.

Installation and Operation

To assist in the reduction of wing flapping and to help in calming the birds, a breast rub must be installed approximately 15 ft. prior to the entrance ramp of the SF-7016 Stunner System and continuing into the Stunner entrance but not interfering with the up-and-down adjustment of the Stunner. It is highly recommended that it be installed beginning at the live hanging conveyor. Simmons recommends the use of indirect, low level lighting for the entrance. Minimize distractions such as reflective surfaces, loud noises or other moving equipment that might cause undue stress.

A foot fogger must also be installed prior to the SF-7016 Stunner System to dampen the feet. This provides not only a better contact with the shackles but also improves the flow of electrical current.

Shackle guide bars must be installed through the Stunner and kept well maintained to allow current to flow through the bird's body. Ensure that shackles are appropriate to the size of the birds being handled. Incorrect fitting shackles are not only painful but may cause an inadequate stun. The time in which the live bird remains on the line ahead of the Stunner should be kept to a minimum and never exceed thirty seconds. This will minimize "pooling" of blood in the wings and any discomfort that may be caused by the shackling of the bird's legs.

2.2.2 Electrical Requirements:

The electrical installation of the SF-7016 Stunner System on-site has been minimized as much as possible due to pre-wiring by the manufacturer.

The following components are pre-wired and mounted on the Stunner Salt Water Control Panel:

- On-Off Switches for the Stunner as well as the water level
- Float switch assembly for water level control
- Solenoid valve with bypass
- Salt water injector assembly

The Stunner Salt Water Control Panel is mounted on the side of the Stunner cabinet or mounted on a wall depending upon your specific situation.

To reduce the risk of possible electric shock, a qualified electrician should perform all necessary wiring.

Installation and Operation

Before installation:

1. Select a suitable location for installation for the Power Supply Enclosure. This should be a well-ventilated, dry location as close to the Stunner installation as possible, ideally within 100 feet (30.48 m).
2. Provide a 15 amp, 115 volts, (220 volts when required), dedicated circuit. The dedicated circuits must not have any other lights, machines or electrical devices connected.
3. Install an electrical conduit, minimum 3/4" (19 mm) diameter from the area of the Power Supply Enclosure Assembly to the area of the Stunner. This conduit will be used to house the three-wire cable, with coax and blue wire from the Enclosure to the Junction box on top of the Stunner.

A factory pre-wired junction box is included in the Stunner Overhead. This box is to be mounted and has red warning lights to indicate when the Stunner is "On". This junction box is connected to the electrical components mounted on the Stunner Salt Water Control Panel. The box has an electrical cable with four conductors and a shielded coaxial cable, attached and coiled. These cables are approximately 100 ft. (30.48 m) long.

Allow enough length to be connected from the Stunner in the Stun/Kill area to the Power Supply Enclosure. If the Stunner will be installed more than 100 ft. (30.48 m) away from the Power Supply Enclosure, notify your Simmons Representative so that longer cables can be supplied.

Note: The Power Supply Enclosure Assembly is a NEMA 4x Stainless steel enclosure. The door is hinged on the left-hand side. The approximate dimensions are 36 - 1/2" (775 mm) high by 24" (610mm) wide by 13" (330 mm) deep. A step-down transformer is provided with the Enclosure for any operation that require 220v.

Installation and Operation

2.2.3. Water Requirements:

A ½-inch (12.7 mm) hose that is supplied with the Stunner System connects the Solenoid Valve on the Stunner directly to the plants water system. To ensure adequate pressure:

- Install a 3/4-inch water line with a ball valve and hose barb to the area of the Stunner.

2.3 Installation and Setup:

Installation of the SF-7016 Stunner should be supervised only by an authorized Simmons technician who is able to provide specific direction to the overall System. This is essential in achieving an optimal installation and start-up for the welfare of the birds and the Stunning System.

During installation, your Simmons Technician will make certain that the SF-7016 Stunning System is properly installed and fully operational, while assuring that all maintenance personnel involved are well instructed in all necessary preventative maintenance, as well as the SF-7016 System's operational functions and conditions.

2.3.1. Cabinet Installation:

1. Position the Overhead Assembly on top of the overhead conveyor at the desired locations ensuring the height adjustment assembly is oriented for easy access.
2. Brace the Overhead Assembly at all four corners using pipe or angle iron, while being sure to keep the assembly level. Braces should be rugged and sturdy enough to prevent rocking of assembly while overhead conveyor and Stunner are in operation.
3. Install the supplied Drop Rods into the lower sleeves of the Overhead assembly and secure them in place using the provided hex bolts, washers and lock nuts. Make sure holes in Drop Rods are all oriented in the same direction (parallel to overhead conveyor).

Installation and Operation

4. Install two guide bars for overhead conveyor shackles. Guide bars should be rigidly braced and mounted parallel to shackles, approximately 3/4" (19 mm) apart, and offset to the breast side so that the bottom of the shackles are approximately 1" (25.4 mm) from center line of the Overhead conveyor.

These bars are a vital part of the stunning operation; serving as the negative contact in conducting the electrical current from the stunning grate through the bird and shackle. Bar ends should be curled and positioned not to get caught in shackles. Shackles should be in constant contact with guide bars through the entire length of the Stunner **(See Figure 1)**.

If Simmons SK-5 Killer is to be installed or is already in place, the Guide Bars should be installed on the breast side of the shackle and should be continuous from the Stunner to the Killer **(See Figure 2)**.

5. With the metal grate in place, install the fiberglass cabinet by sliding upwards onto Drop Rods and positioned approximately 7"- 9" (178 mm – 229 mm) from grate to bottom of shackle. Making sure to keep the fiberglass cabinet level in all directions, select the correct hole in the cabinet leg and secure each leg to the Drop Rod using the provided hex head cap screws and locknuts.
6. Install 3/4" (19 mm) water supply line with valve and 3/4" (19 mm) x 1/2" (12.7 mm) reducer bushing. Using the fittings and tubing furnished with the Stunner, connect the Salt water Injector System as follows:

Step 1) Install male 1/2" (12.7 mm) hose barb x 1/2" (12.7 mm) hose to plant water line.

Step 2) Install 1/2" (12.7 mm) hose from barb to 1/2" (12.7 mm) Water Valve assembly.

Installation and Operation

2.3.2. Electrical Installation:

The DC Power Supply and AC power supply are similar in appearance, but **ARE EACH DESIGNED FOR A SPECIFIC FUNCTION. THEY CAN ONLY BE USED FOR THEIR INTENDED PURPOSE.**

Failure to heed this warning may cause an inferior product, endanger safety and damage the units, thus voiding all warranties. Most importantly, this would be detrimental to the comfort and welfare of the birds involved. Simmons Engineering is not responsible for the consequences resulting from any unauthorized use of its products. To reduce the risk of possible electric shock, wiring should be performed by a qualified electrician.

1. Mount the Power Supply Enclosure Assembly. This should be in a well-ventilated, dry location as close to the Stunner installation as practical, within 100 feet (30.48 m) if possible.
2. Install the Junction box on the Overhead Assembly.
3. The Power Supply Enclosure comes complete with Primary and Standby units. The Enclosure is connected on a dedicated circuit. Install wiring as shown in **Figure 3**.
4. At the Junction Control Box mounted on the Stunner Salt Water Control Panel, turn the Water Switch (Black Switch) "ON". This will activate the Float Switch and Solenoid Valve, and turn water on to the Stunner cabinet.
5. At the Junction Control Box mounted on the Stunner Salt Water Control Panel, turn Stunner Switch (Red Switch) "ON". This will energize the circuit to activate the Power Supply Enclosure.
6. Inside the Power Supply Enclosure, set the Primary / Standby Selection Switch to "Primary".
7. Turn the DC Power Supply switches to "ON". The output meter selector switches to "VOLTS" and monitors the voltage output on the Primary Supply's digital meter, while turning the Primary Supply's output level control from 0 to full scale. Verify the readings are between 0 and approximately 45 volts. 0 volts with 0.2 to 0.3 amps and no variation with the output level control, indicates a short circuit from the Power Supply Enclosure to the Stunner contactor grate.
8. Set the Stunner Selection Switch to "STANDBY" and repeat Step 7 as mentioned above for the Standby Stunner Power Supply. Return the selectors switch to the "PRIMARY" position.

Installation and Operation

9. Using a 0 to 50 DC Voltmeter, test the voltage between the primary Stunner Power Supply Enclosure red wire (+) and the green/white wire (-). Then test the voltage between the contactor grate (+) and the shackles guide bar (-) and compare the reading. If the coax cable is greater than 100 feet in length, there may be a slight voltage difference. If the voltage is 0, or very low, at the cabinet with the Stunner output control fully clockwise, there is an open circuit between the Control Enclosure and the Cabinet.

If the System fails to power up:

- o Turn off the primary or standby control and reset
- o Check incoming power to the system
- o Check the fuse
- o Switch to Standby Control and repeat start procedures

Warning: Ensure that there is no possibility of a direct short between the contactor grate and ground at any time. A short would cause serious damage to the Stunner Power Supplies. Attempting to repair the Simmons Stunner Power Supplies or attempting to open the case in any way VOIDS ANY AND ALL WARRANTIES. Please contact Simmons for any and all service.

Installation and Operation

2.4. How to Operate:

Before every operation of the System, the following procedures should be checked for the welfare of the birds and the SF-7016 Stunning System.

1. Fill the cylinder of the Salt water Injector with salt; food-grade salt pellets are recommended for the best performance.
2. At the Control Box on the Stunner Salt Water Control Panel, turn Water Switch (BLACK Switch) "ON".
3. At the Control Box on the Stunner Salt Water Control Panel, turn Stunner Switch (RED Switch) "ON".
4. At the Stunner Power Supply Enclosure, with the Primary/Standby Selector Switch in the "PRIMARY" Position, set the output level control fully counterclockwise.
5. Turn the primary Stunner Control Unit power "ON":
 - Digital meter comes on
 - Overhead warning light illuminates
6. Set the meter selector switch to "VOLTS," and turn the output-level control for a reading of 35 - 45. The Stunner Power Supply is now ready for operation, and the contactor grate is charged so that incoming birds will be stunned. Adjust the output level control setting for the installation. Set the meter selector switch to "AMPS" to monitor the current loading on the contactor grate.
7. To establish the correct voltage for stunning, lift and shackle birds securely by both legs. Ensure that shackles are appropriate to the size of the birds being handled. Incorrect fitting shackles are not only painful but may cause an inadequate stun. The time in which the live bird remains on the line ahead of the Stunner should be thirty seconds or less. Do not exceed this limit. Staying within the allotted time will minimize "pooling" of blood in the wings and any discomfort that may be caused by the shackling of the bird's legs. Do not shackle birds with broken legs or wings as they should be immediately killed humanely rather than shackled.

Installation and Operation

8. The height of the Stunner is as important as the voltage and amperage settings.
- Only the head and neck of the birds should be in contact with the grate. Adjust the height of the cabinet so that the distance from the bottom of the shackle to the contactor grate in the bottom of the cabinet is the suggested height (Height parameters in **Figure 2**). The head and neck of the birds should be in contact with both the water and the grate.
 - The height of the cabinet should be adjusted to allow the feathers on the head and upper neck of each bird to become lightly wetted with saline and prepared for electrical contact to the contactor grate. The anesthetizing current generated by the Stunner Power Supplies, flows from the contactor grate through the bird to the shackle and into the shackle guide bar, which is connected by the ground wire to the Stunner power controls. With a measured, adjustable voltage level, the stunning is instantaneous and humanely anesthetizes the bird.

ABSOLUTELY NO WINGS OR SHOULDERS SHOULD CONTACT THE WATER IN ORDER TO AVOID DAMAGE.

9. Turn on water supply to the Solenoid Valve and set the water level. Turn Water Level Switch "ON" and water should begin flowing through the Solenoid Valve. Adjust the water level by sliding the Switch Assembly up and down in its housing. When the desired water level is reached, tighten the clamp bolt on the Switch Assembly. This will maintain the correct water level. The entire Switch Assembly can be lifted out of the housing for cleaning and maintenance.
10. The cylinder for the Salt water Injector Assembly should be filled each day with salt pellets and replenished as needed. The salt cylinder will likely require one refill of salt pellets during an eight-hour shift. If more salt is needed, the saline level may be too high and overflowing from the Stunner cabinet, or the incoming water pressure may be too high. Slowly adjust the saline level sensor position with short trial adjustments until the saline solution reaches the correct depth above the contactor grate, but not overflowing.

Do not allow any water to overflow from the Stunner to the floor. This would act just like a copper wire and would bleed voltage away from the Stunner, causing the power setting to increase.

Installation and Operation

Warning:  Indicates hazardous voltages may be present. Use standard safety precautions to avoid personal contact with these voltages. For maximum safety, DO NOT TOUCH this System while in operation. Do not touch any object that could provide a current path to the System. Before performing any maintenance, use your facility's standard operating procedures and lock out the System.

3.1 Daily Prior to Start-Up:



- Lower the grating from the upright position to the horizontal position.
- Turn on DC and AC Power Supplies and set the anticipated voltage.
- Use a handheld meter to verify the voltage on both the AC and DC plates.
- Raise or lower the stunner cabinet to the desired height via your manual, electric, hydraulic or pneumatic lift system.
- Verify that the red warning light is functioning above the stunner.
- Verify that the water is on and the salt dispenser is full.
 - o The salt cylinder will likely require one refill of salt pellets during an eight-hour shift.
 - o If more salt is needed, the saline level may be too high and overflowing from the Stunner cabinet, or the incoming water pressure may be too high. Slowly adjust the saline level sensor position with short trial adjustments until the saline solution reaches the correct depth above the contactor grate, but not overflowing.

Maintenance and Cleaning Instructions

3.2 Maintenance and Cleaning Instructions:

Maintenance and cleaning of the SF-7016 Stunning System is a vital step in ensuring that the System provides an excellent stun for the welfare of the birds and the continued performance of the System. Please refer to the Parts Manual for detailed instructions on replacement parts and contact Simmons Engineering for authorized parts.

3.3 Daily Maintenance / Sanitation:

DO NOT BEGIN MAINTENANCE / SANITATION PROCEDURES UNTIL THE SYSTEM IS CONFIRMED TO BE IN THE “OFF” POSITION.

- Ensure that the SF-7016 cabinet, including the grating are cleaned and sanitized daily.
 - o During sanitation, lift the grate sections and latch with the grate clips.
 - o Clean and sanitize both the top and bottom side of the grate to remove any salt buildup that may have occurred during production.
 - o Leave the grating in the vertical position for a period to allow any excess water and salt to drain from the grate.
 - o Oil the hinge on the stunner grates to ensure that any excess salt buildup is properly removed from the hinge.
- With the controls in the “OFF” position, inspect all control dials to ensure that they are free from buildup and operating properly.
 - o Switch between the “Primary” and “Backup” on the Power Supplies daily to ensure that both supplies are functioning normally.
 - o If you notice a voltage reading on the DC supply that is significantly higher than normal, you may have a damaged Power Supply. Normal maximum voltage readings should be no greater than 50 volts.
- Adjust the height of the cabinet up and down using your specific lift system (manual, electric, hydraulic or pneumatic) up and down to ensure that the lift is operating freely and normally.

Section III - Maintenance and Cleaning Instructions

Maintenance and Cleaning Instructions

3.4 Weekly Maintenance / Sanitation:

- Check the stainless-steel guide bars for proper adjustment.
- For stainless steel shackles:
 - o Ensure that the guide bars are in constant contact with the stainless-steel shackles.
- For plastic shackles:
 - o Ensure that that the guide bars are in constant contact with the knee area of the bird.
- Check the stainless-steel chain on the overhead lift mechanism. Ensure that it is tight and not slipping on any of the sprockets. Tighten any assemblies as needed.
- Remove the float switch from the housing and clean any accumulated dirt and buildup around the floor.

Section III - Maintenance and Cleaning Instructions

Maintenance and Cleaning Instructions

3.5 Monthly Maintenance:

- Visually inspect the stainless-steel chain on the overhead system for excessive wear.
- Visually inspect the sprockets on the overhead chain while in operation to ensure that it is functioning properly and that the chain is tight.
- Visually inspect the grate wiring connections and ensure that they are tight and not showing any signs of corrosion.
- Turn off the water and remove one of the hoses from the bottom of the salt dispenser. Allow water to drain.
 - o Open the lid on the dispenser and rinse the inside of the canister to remove any salt buildup that may have occurred.
 - o Remove the salt meter from the base of the dispenser and clean the salt injection tubes. Reattach the meter to the salt dispenser.
 - o Reattach the tube to the bottom of the dispenser and turn the water to check for leaks.
- Check the pneumatic air/water separator and blow out any excess water.
- Check the oiler to ensure the proper oil level is maintained.

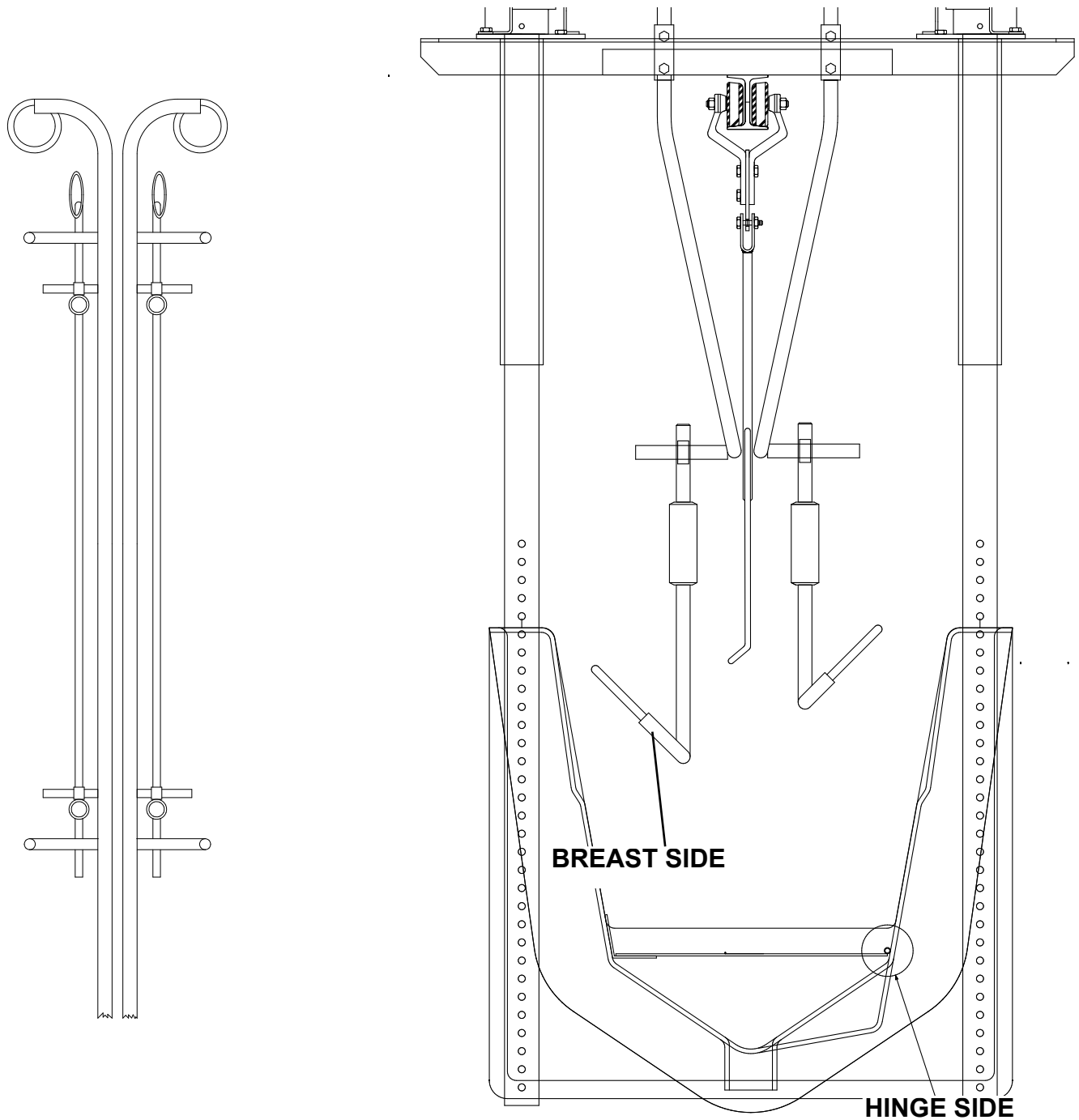
3.6 Quarterly Maintenance:

- Grease all sprockets on the overhead using the supplied grease fittings.
- Check all the bolts in the system to ensure they are properly secured.

Troubleshooting the System

PROBLEM	POSSIBLE CAUSE	SOLUTION
BIRDS NOT FEEDING INTO KILLER	Birds not stunned properly.	- Check Stunner voltage. - Check shackle guide bars in Stunner to be sure shackles are in constant contact with guide bars. - Verify Salt System is working correctly - Verify that water is not overflowing from Stunner in a steady stream and "grounding" the Stunner.
SUSPECTED STUNNER DAMAGE TO BIRDS	Voltage is too high or birds "pogoing" through stunner.	- Check voltage. - Check shackle guide bars.
SHORT BIRDS NOT BEING STUNNED	Stunner height set too low.	- Raise height of cabinet.
SALT INJECTION SYSTEM NOT WORKING PROPERLY	Metering Block may be plugged. Solenoid bypass valve may be open.	- Disassemble and clean. - If Bypass Valve is left open so that solenoid cannot "pulse" on and off, saline solution will not be pulled from cylinder. Close Bypass Valve.
WATER LEVEL CONTROL NOT WORKING	Float Switch may be jammed. Solenoid Coil may be bad. Transformer may be bad.	- Lift Float Switch out and clean well. - Replace Coil if needed. - Check output voltage at terminals or at solenoid. Should be 24V AC. Replace if needed.

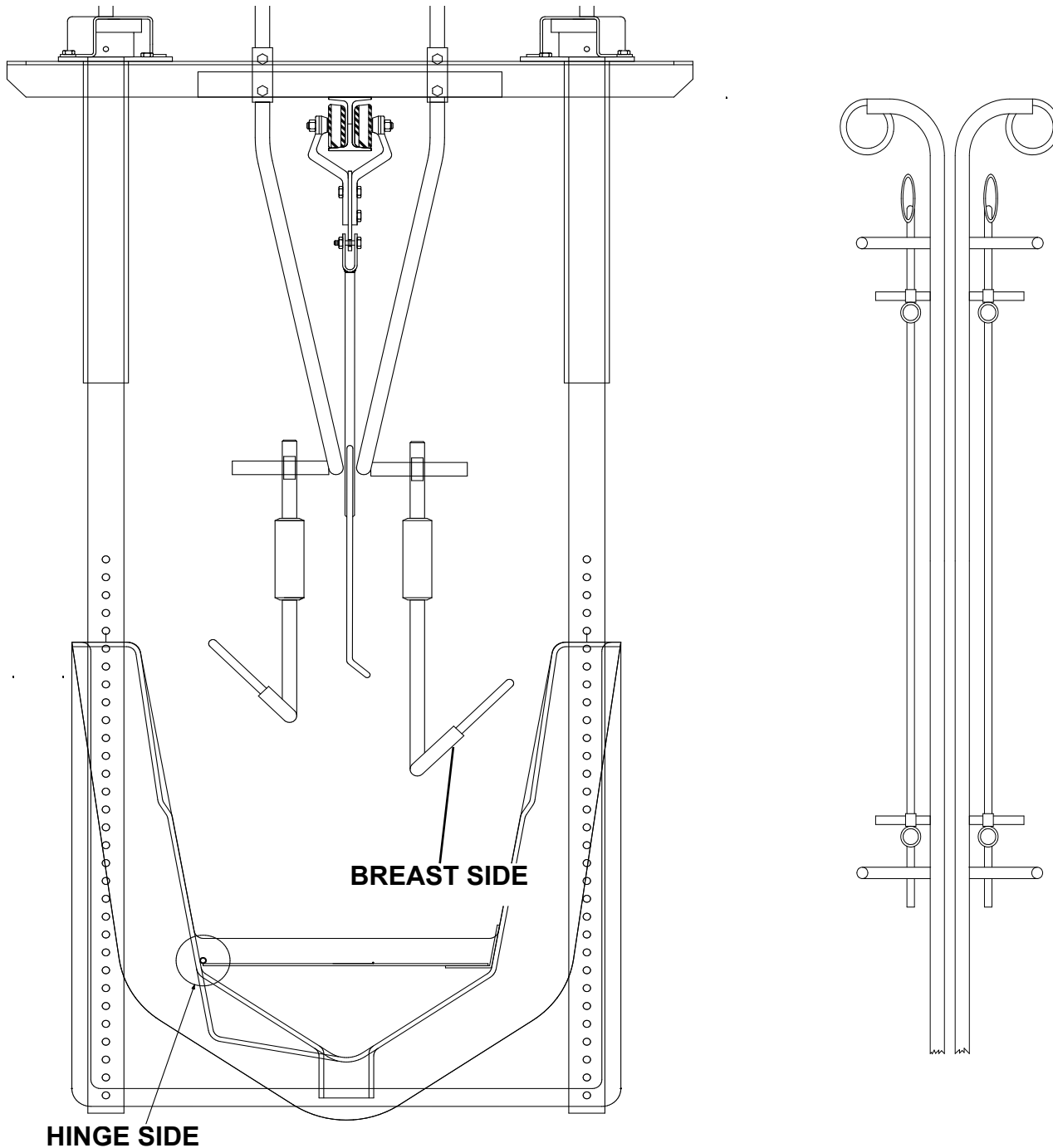
**Stunner Guide Bar Installation
(Figure 1L)**



LEFT HAND STUNNER

VIEW IS FROM ENTRANCE END OF STUNNER

**Stunner Guide Bar Installation
(Figure 1R)**

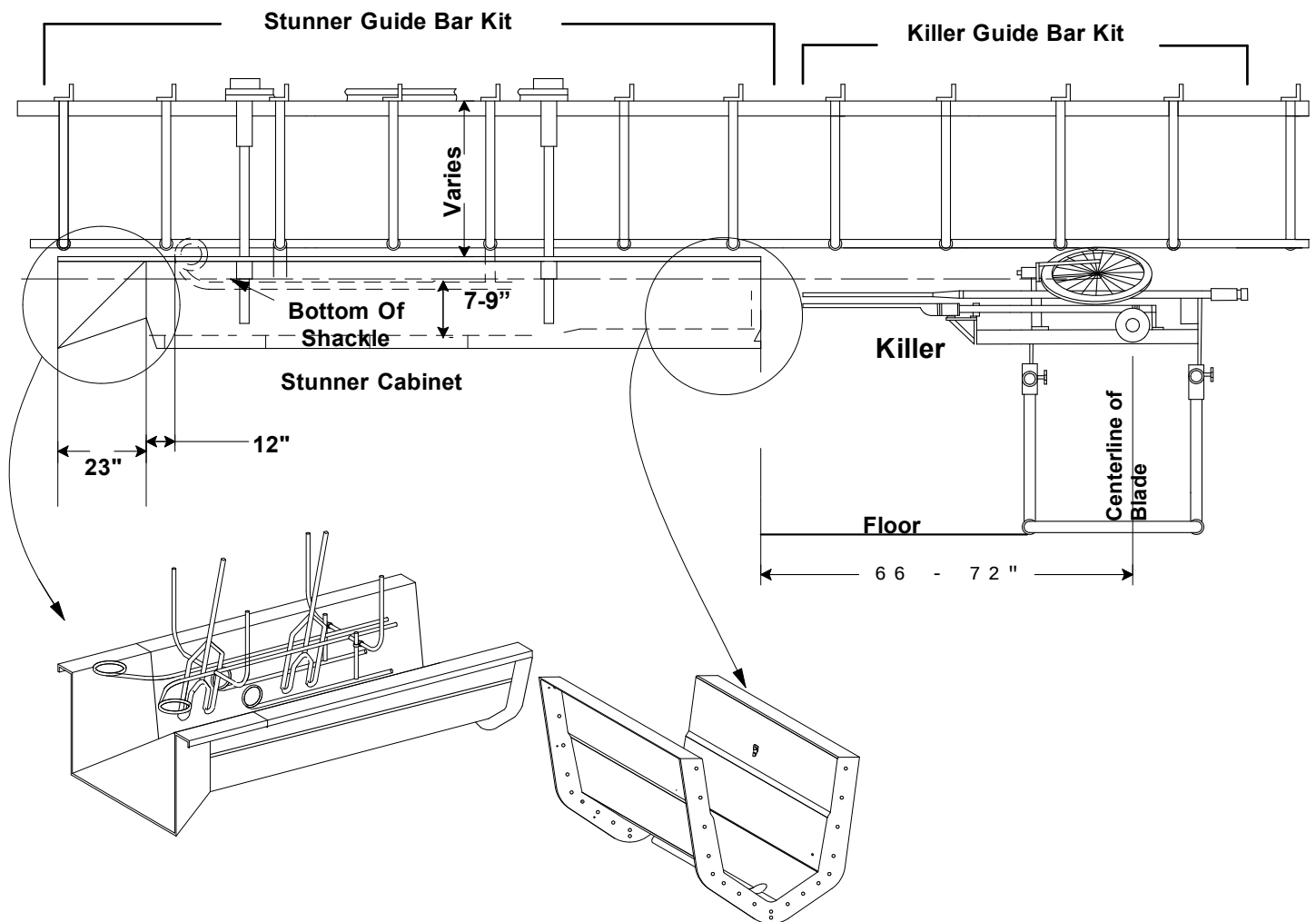


RIGHT HAND STUNNER

VIEW IS FROM ENTRANCE END OF STUNNER

Figure 2 - Stunner & Killer Guide Bar Installation

**Stunner & Killer Guide Bar Installation
(Figure 2)**



NOTE: Breast side Guide Bar continues to SK-5 Killer Entrance

Figure 3 - Power Supply Enclosure Wiring Diagram

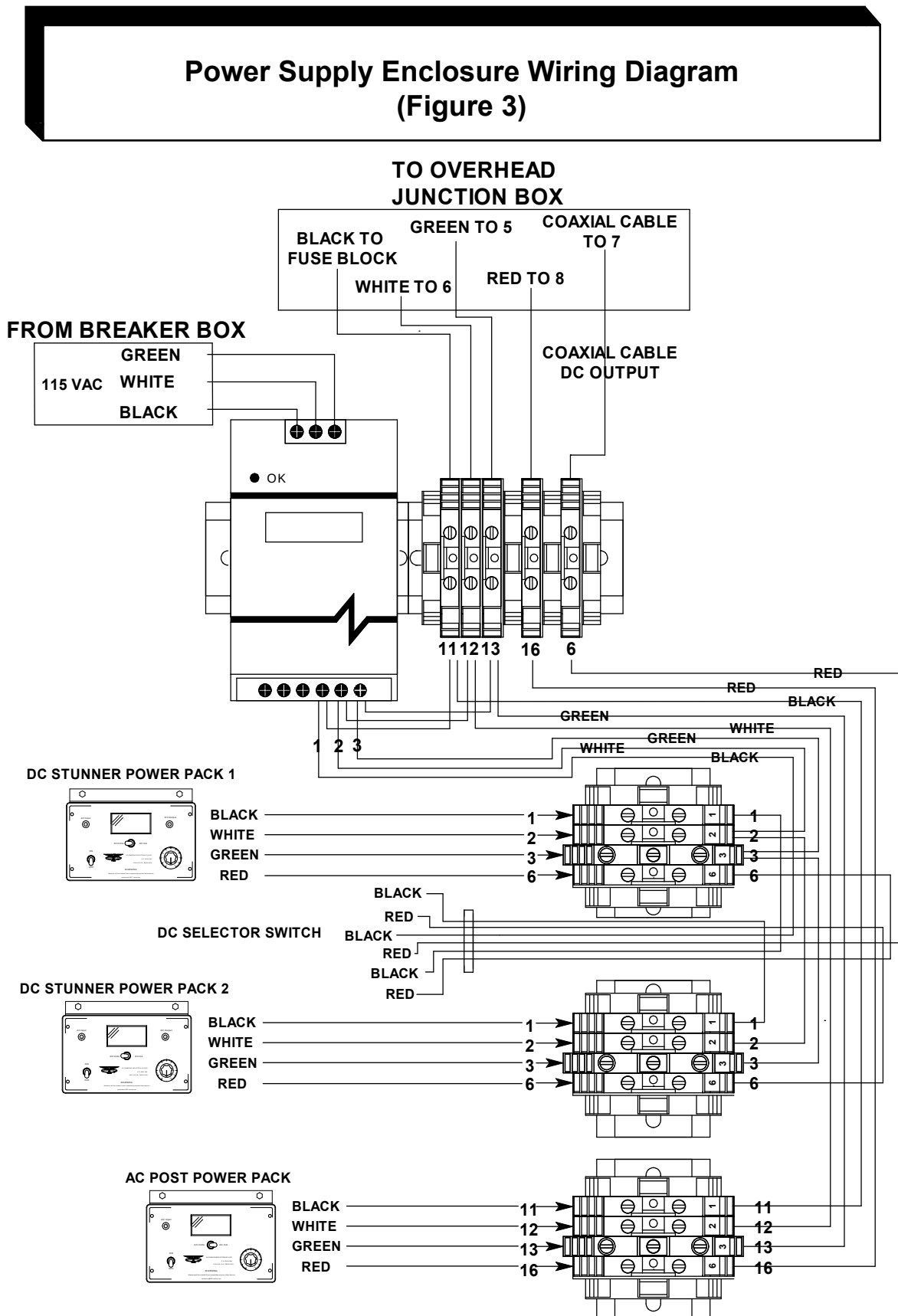


Figure 4 - Overhead Junction Box Wiring Diagram

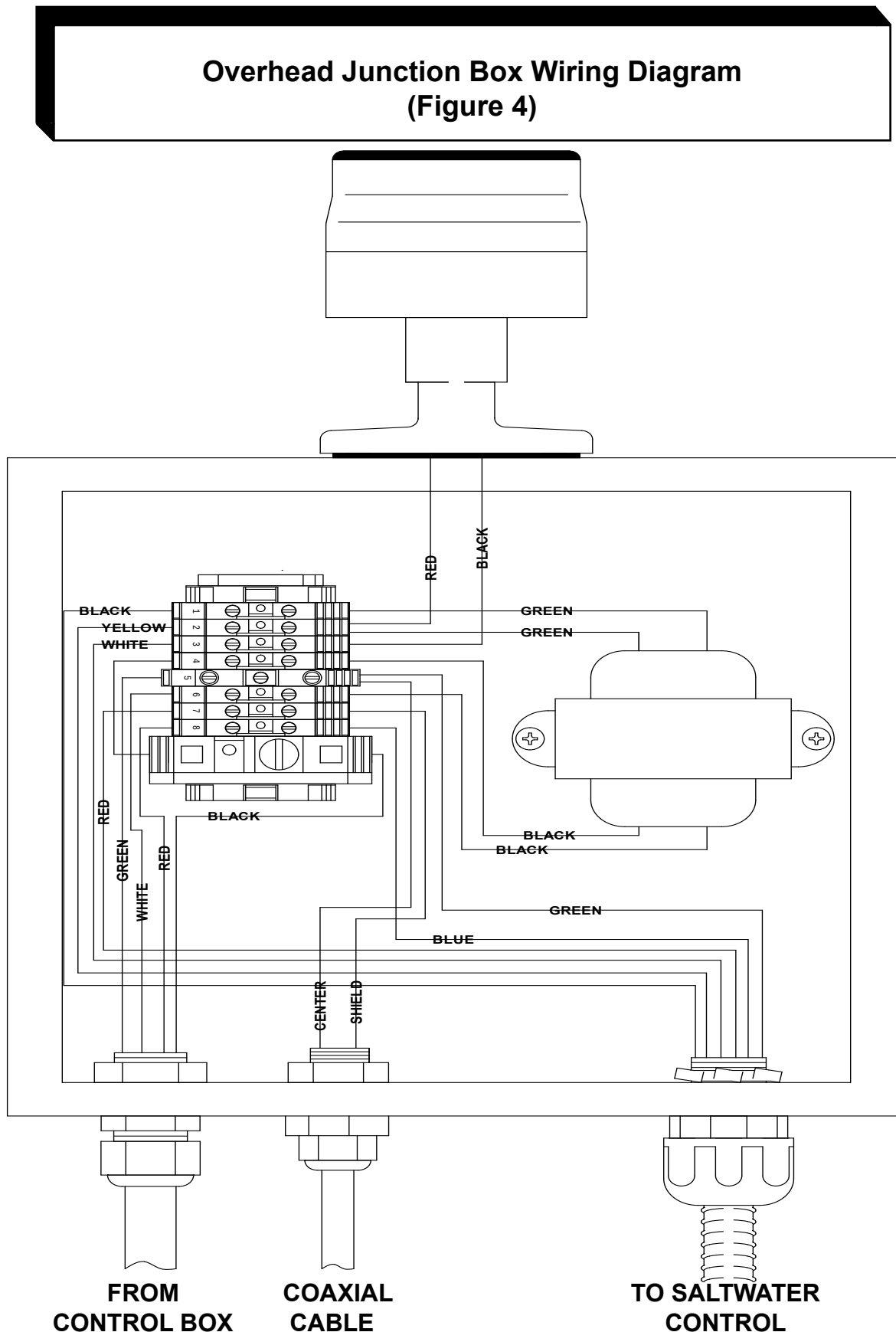


Figure 5 - Stunner Control Box Wiring Diagram

**Stunner Control Box Wiring Diagram
(Figure 5)**

